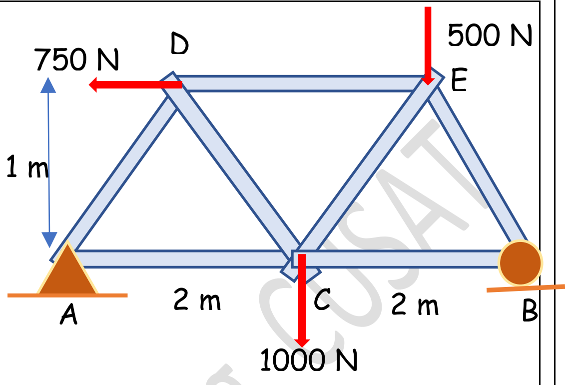
ANALYSIS OF TRUSSES BY METHOD OF SECTIONS

We have leant to analyse the forces in a plane truss using the method of joints. In that we have considered the equilibrium of each joint. The free body diagram of each joint consists of a system of concurrent forces. We can use two equilibrium equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$. Hence for solving the problem easily, we begin with a joint with maximum two unknown forces.

#Example 1

Let us consider a plane truss as shown. Instead of isolating a joint, if we cut through the whole truss, separate the halves, and consider the free body diagram of any one part, we will get a system of general case of forces. Then we can use three equilibrium equations $\Sigma F_x = 0$ $\Sigma F_y = 0$, and $\Sigma M = 0$. Now we can find out three unknowns from these equations.



If we carefully take sections in such a way that only a maximum of three unknown force members are cut, the solution procedure becomes easier.

Let us find out the support reactions first by considering the free body diagram of the complete truss.

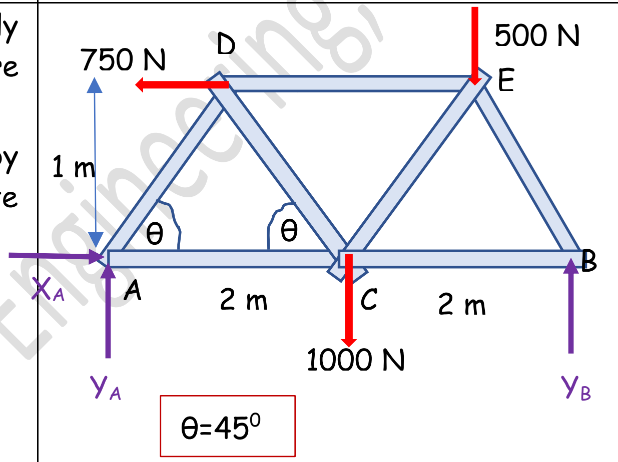
$\Sigma F_x = 0$ ⟶ $X_A - 750\,N = 0$ ⟶ $X_A = 750\,N$

$\Sigma F_y = 0$ ⟶ $Y_A + Y_B - 500\,N - 1000\,N = 0$

$\Sigma M_A = 0$

$Y_B \times 4m - 1000\,N \times 2m - 500\,N \times 3m + 750\,N \times 1m = 0$

$Y_B = 687.5\,N$ $\quad$ $Y_A = 812.5\,N$



Cut the members AC, DC, and DE and separate the two parts. Consider the equilibrium of left part.

$\Sigma M_D = 0$

$X_A \times 1m - Y_A \times 1m + F_1 \times 1m = 0$

$F1 = 62.5\,N$

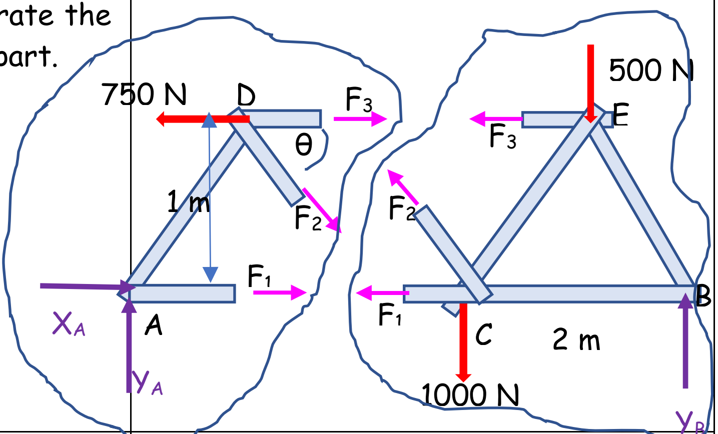
$\Sigma F_y = 0$

$Y_A - F_2 \sin 45 = 0$ ⟶ $F_2 = 1148.3\,N$

$\Sigma F_x = 0$

$X_A + F_1 + F_3 + F_2 \cos 45 - 750 = 0$

$F_3 = -874.5\,N$



Now, cut the members AD, DC, CE, and CB and consider the equilibrium of one of the parts.

$\Sigma M_C = 0$ ⟶ $-Y_A \times 2m - F_4 \sin 45 \times 2m = 0$

$F_4 = -1149.1\,N$

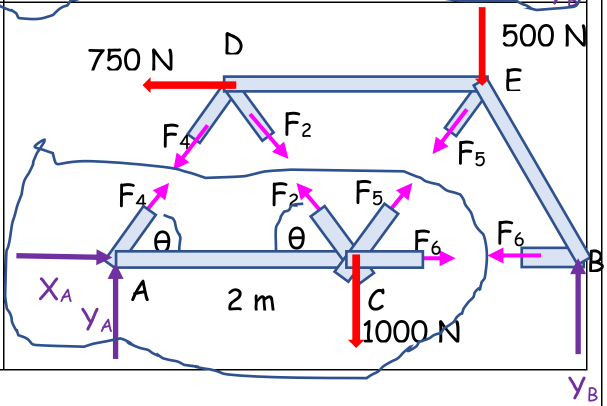
$\Sigma F_y = 0$ ⟶ $Y_A + F_2 \sin 45 + F_4 \sin 45 + F_5 \sin 45 - 1000 = 0$

$F_5 = 266\,N$

$\Sigma F_x = 0$

$X_A + F_4 \cos 45 + F_6 - F_2 \cos 45 + F_5 \cos 45 = 0$

$F_6 = 686.4\,N$



We can now section members AD, DC, CE, and BE and consider the equilibrium of bottom part.

$\Sigma F_x = 0$

$X_A + F_4 \cos 45 - F_2 \cos 45 + F_5 \cos 45 - F_7 cos45 = 0$

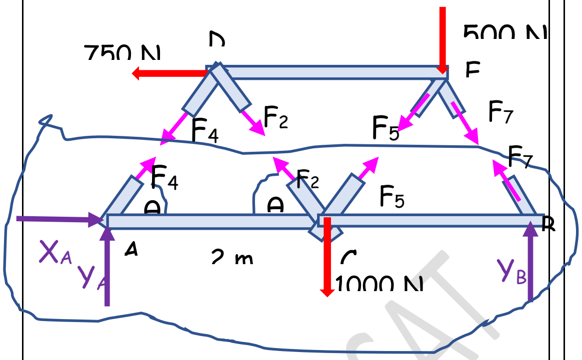
$F_7 = -970.7\ N$

Check!!!!

$\Sigma F_y = 0$

$Y_A + F_2 \sin 45 + F_4 \sin 45 + F_5 \sin 45 - 1000 + Y_B + F_7 sin\ 45 = 0$

$Y_B = 687\ N$

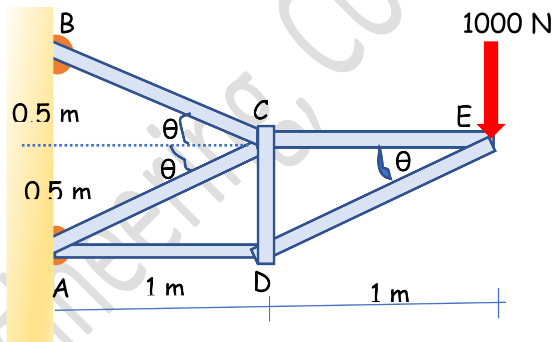


#Example 2

Calculate the forces in the members of the truss shown by method of sections.

Since the truss is a cantilever type, we need not find out the support reactions. We can take sections of the truss and consider the equilibrium of free end part.

tan θ=0.5 $\boxed{\Theta = 26.6^0}$



We cut members AD, DC, and CE and consider the equilibrium of the right part.

$\Sigma M_D = 0 \Rightarrow \quad -1000 \times 1m + F_3 \times 0.5m = 0$

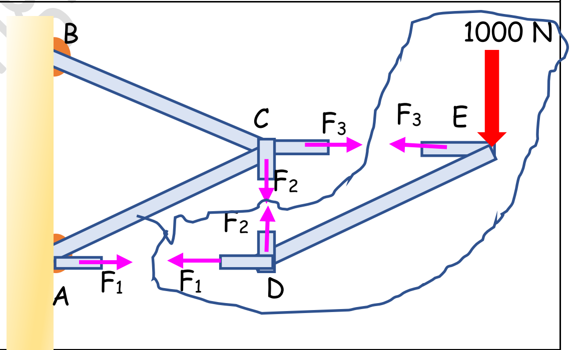
$F_3 = 2000\ N$

$\Sigma F_y = 0 \Rightarrow \quad F_2 - 1000\ N = 0$

$F_2 = 1000\ N$

$\Sigma F_x = 0 \Rightarrow \quad -F_1 - F_3 = 0$

$F_1 = -2000\ N$



A new section is considered by cutting members CA, CB, CD, and DE

$\Sigma M_C = 0 \Rightarrow \quad -1000 \times 1m - F_6 \sin 26.6 \times 1m = 0$

$F_6 = -2236\ N$

$\Sigma F_x = 0 \Rightarrow \quad -F_4 \cos 26.6 - F_5 \cos 26.6 - F_6 \cos 26.6 = 0$

$-F_4 - F_5 = -2236$

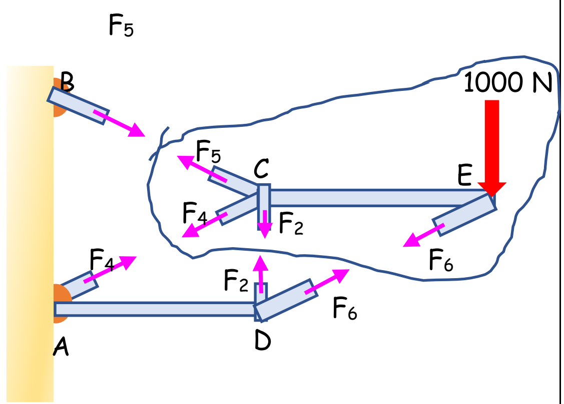
$\Sigma F_y = 0 \Rightarrow \quad -F_2 - F_4 \sin 26.6 + F_5 \sin 26.6 - F_6 \sin 26.6 - 1000 = 0$

$-1000 - F_4 \sin 26.6 + F_5 \sin 26.6 - (-2236) \sin 26.6 - 1000 = 0$

$-F_4 + F_5 = 2236$

$F_4 = 0$

$F_5 = 2236\ N$

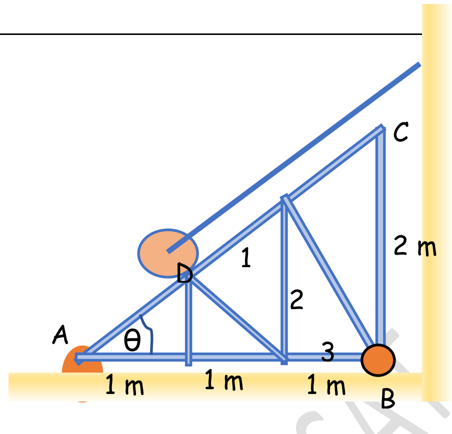


#Example 3

Using the method of section, find the force produced in bars 1, 2, and 3 of the trusses shown due to the action of a ball of weight W= 1000 N and radius R=0.5 m resting on the hinge D and supported by a cable as shown

$\tan\theta = \frac{2}{3}$ ⟶ θ =33.7^0

Let us consider the free body diagram of the whole truss. The force acting at joint D is the action of the ball of weight W.



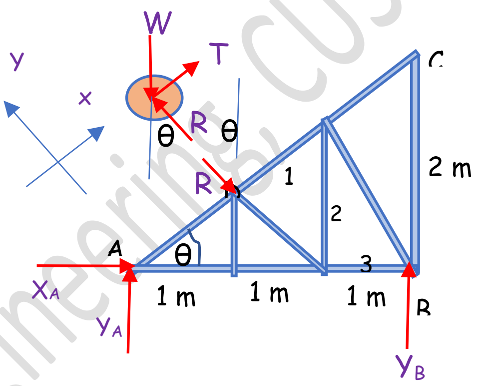
Consider the free body diagram of the ball. We are interested only in the force exerted by the ball on the truss. Hence, we can resolve the forces parallel and perpendicular to AC.

$\Sigma F_y = 0$ ⟶ $R - W\cos 33.7 = 0$

$R = 1000\cos 33.7 = 832\ N$

$$\cos\theta = \frac{1\ m}{AD}$$

$$AD = \frac{1\ m}{\cos 33.7} = 1.2\ m$$



Now consider the free-body diagram of the whole truss.

$\Sigma F_x = 0$: $X_A + R\sin 33.7 = 0$ ⟶ $X_A = -832\sin 33.7 = -461.6\ N$

$\Sigma M_A = 0$: $Y_B \times 3m - R\times AD = 0$ $\quad Y_B = \frac{832\times 1.2}{3} = 332.8\ N$

$\Sigma F_y = 0$: $Y_A + Y_B - R\times\cos 33.7 = 0$ $\quad Y_A = 359.4\ N$

Consider a section cutting through members 1, 2, and 3 of the truss and the equilibrium of the left part

$\Sigma M_A = 0$: $F_2 \times 2m - R\times AD = 0$: $F_2 = 499.2\ N$

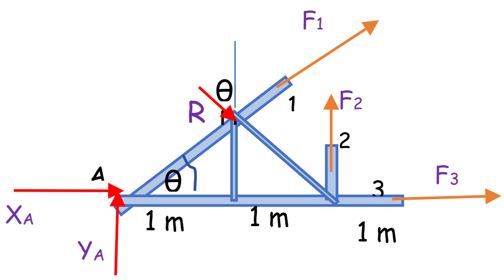
$\Sigma F_y = 0$: $Y_A + F_2 - R\times\cos 33.7 + F_1 \times\sin 33.7 = 0$

$F_1 = -300\ N$

$\Sigma F_x = 0$

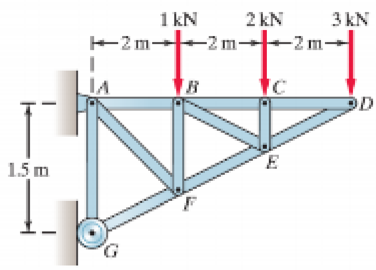
$X_A + F_1\cos 33.7 + F_3 + R\sin 33.7 = 0$

$F_3 = 250\ N$



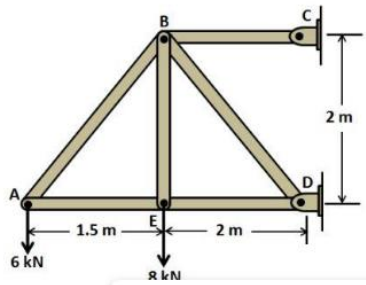
#Exercise 16

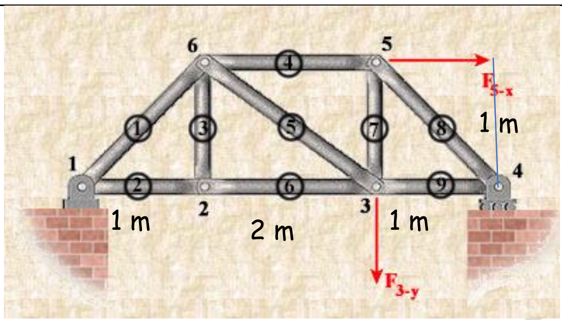
16.1 Find the forces in all the bars of the truss



16.2

Find the forces in all the bars of the truss



16.3 Find the forces in all the bars of the truss, if F_{5-x}= 1000 N, F_{3-y}= 750 N	
16.4 Find the forces in the bars DE, EF, and FG of the truss